Foundation Model Insights and a Multi-Model Approach for Superior Fine-Grained One-shot Subset Selection

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Why We Study One-Shot Subset Selection

- We see that training on full datasets is costly in compute, time, and energy.
- We can preserve accuracy with smaller, well-chosen subsets while cutting cost.
- We leverage Foundation Models (FMs) for strong, general-purpose representations without task-specific pretraining.

Our Problem: One-Shot Subset Selection with Minimal Compute

- Our goal: we pick a small, informative subset once (no iterative retraining).
- We note that traditional pipelines rely on dataset-pretrained IEs, which are costly and dataset-dependent.
- We ask: Do FMs improve selection, and are all FMs equally good as IEs?

From Dataset-Dependent IEs to Foundation Models

- We replace task-specific IEs with FMs to remove dataset dependence.
- We investigate when FMs help and which FM we should use.

Figure: One-shot subset selection: traditional IE vs. a single FM. Our FM-based IE excels on fine-grained data.

What We Show and Contribute

- We find that FMs shine on fine-grained data; their advantage fades on noisy, coarse-grained data.
- We show that higher FM task accuracy does not imply better subset selection.
- We propose RAM-APL, which fuses intra-class prototypicality (RAM) and inter-class ambiguity (APL) with adaptive weighting.
- We achieve state-of-the-art results on Pet, Food-101, and CUB-200-2011.

Method: RAM-APL Pipeline at a Glance

- We extract features per sample from multiple FMs (no target-dataset finetuning).
- We compute RAM: we rank within each class by distance to the class center per FM; we then average normalized ranks across FMs.
- We compute APL: we use pseudo-label correctness across FMs to capture inter-class ambiguity.
- We fuse scores with sampling-rate-dependent weights and we select the smallest fused scores.

RAM: Intra-class Prototypicality via Ranking

- We compute per-FM class centers and rank samples by distance (closer means more prototypical).
- We normalize ranks to align unaligned feature spaces across FMs.
- We average ranks across FMs to obtain a robust prototypicality score; lower scores indicate more representative samples.

RAM in Action: What Low vs. High RAM Looks Like

- We observe that low-RAM samples have clear objects, simple backgrounds, and prototypical views.
- We observe that high-RAM samples exhibit atypical poses, occlusions, or cluttered backgrounds.

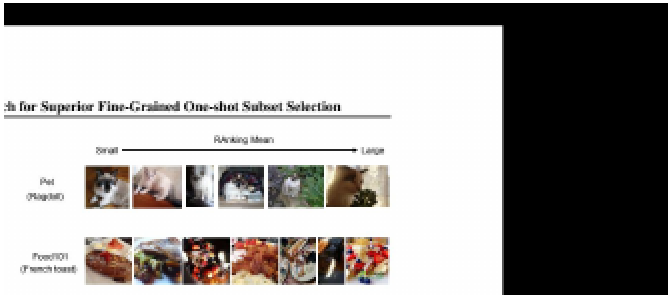

Figure: Qualitative examples illustrating the RAM metric.

APL: Inter-class Ambiguity via Pseudo-label Accuracy

- We assign a pseudo-class by nearest class center for each FM.
- We mark correctness against ground truth and average across FMs.
- We treat lower APL (more confusion) as flags for discriminative, boundary samples.

Adaptive Fusion: Curriculum over Sampling Rate

- Score = W1 × RAM + W2 × (1 − APL).
- At low budget p, we prioritize W1 (representative/easy).
- As p grows, we increase W2 to include more discriminative/hard samples.
- By default, we use alpha 0.2 and beta 1 to provide robust weighting across datasets.

Why Multi-FM? Complementarity of Representations

- We find that different FMs encode diverse semantics; naïve "best FM" heuristics fail.
- We observe low cosine similarity across FMs, which justifies fusion over a single FM.
- We use ranking-space fusion to avoid alignment and concatenation pitfalls.

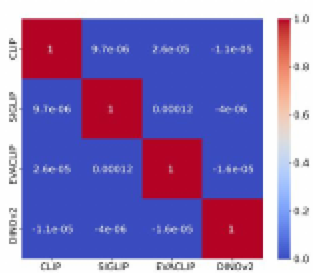

Figure: Cosine similarity matrix across FM embeddings.

Experimental Setup: Datasets, Baselines, Protocol

- We evaluate on CIFAR-10/100, CIFAR-10N/10I, Oxford-IIIT Pet (and a noisy variant), CUB-200-2011, and Food-101.
- We compare to baselines: diversity—K-Center Greedy/CoreSet [Sener18], MDS (Kennard–Stone) [Kennard69], CD (Cluster-Distance, k-means) [MacQueen67]; graph-cut—GC [Wei13]; uncertainty—Margin [Settles10]; optimization—GraNd [Paul21], GLISTER [Killamsetty21], CRAIG [Mirzasoleiman20], CAL (class-balanced CRAIG) [Mirzasoleiman20]; herding—Herding [Chen12]; prototype—MIN (Nearest-Prototype; prototypical proximity) [Snell17]; plus Random.
- We use MobileNet-V3 as one of the target models and evaluate across multiple sampling rates.
- We conduct a Single-Model Study and multi-model (RAM-APL) experiments.

Single-Model Study Protocol

- We compare three IE types: model-TD (target-pretrained), model-TIN, and FMs.
- We vary selection algorithms and sampling rates.
- We diagnose when FMs help and which FM is best for selection.

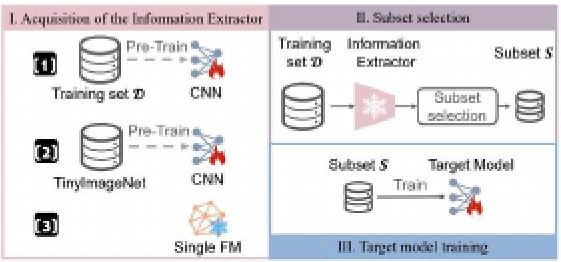

Figure: Framework of our Single-Model Study.

Finding 1: Fine-Grained Gains; Coarse/Noisy Are Mixed

- On fine-grained datasets (e.g., Pet, CUB-200-2011), FMs are frequently optimal across budgets and selectors.
- On clean but coarse-grained data (e.g., CIFAR-100), FM advantage is less consistent.
- Under label noise, FMs remain strong for fine-grained Pet but offer limited gains on noisy coarse-grained CIFAR-10.

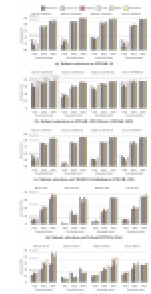

Figure: FM-based selection excels on fine-grained tasks; benefits are mixed for coarse or noisy data.

Finding 2: FM Task Accuracy Does Not Imply Selection Quality

- We find that higher FM zero-shot or finetuned accuracy does not ensure better selection.
- We observe that different FMs win under different budgets and selection algorithms.
- We are motivated to use multi-FM fusion rather than pick a single "best" FM.

Figure: FM target accuracy does not equal subset selection quality across models.

We Outperform Baselines on Fine-Grained Datasets

- We achieve consistent gains over Random, GC [Wei13], MDS [Kennard69], and KCG [Sener18] across budgets.
- We evaluate on Oxford-IIIT Pet, Food-101, and CUB-200-2011.
- We find that adaptive fusion selects representative first and then discriminative samples effectively.

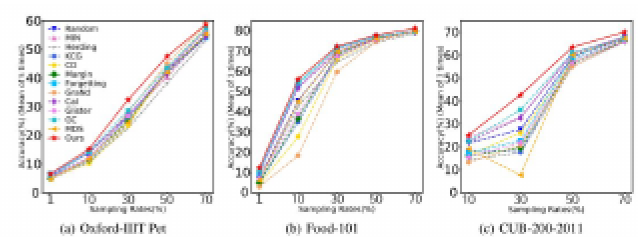

Figure: Comparison with baselines on three classic fine-grained datasets (average accuracy).

Baselines: Herding [Chen12]; K–Center Greedy/CoreSet [Sener18]; GC (Graph Cut) [Wei13]; MDS (Kennard–Stone) [Kennard69]; CD (k-means cluster distance) [MacQueen67]; Margin [Settles10]; Forgetting [Toneva19]; GraNd [Paul21]; GLISTER [Killamsetty21]; CRAIG

Ablation: RAM vs. RAM-APL vs. MIN (Pet) — Mandatory Table 1

- We find that RAM (multi-FM ranks) already surpasses single-FM MIN [Snell17].
- We observe that adding APL further improves performance, especially at higher budgets.
- We conclude that combining intra-class and inter-class signals helps.

Table: Table 1. Ablation study on Pet. "Model-TD" refers to the model pretrained on Pet for 10 epochs. Values are Top-1 accuracy (mean±std).

Method	IE	1%	50%	70%
Model-TD	Model-TD	5.6±0.7	40.3±2.6	55.2±2.7
MIN [Snell17]	CLIP	5.6±0.2	45.9±1.8	56.3±0.7
MIN [Snell17]	DINOv2	6.2±0.1	46.8±2.0	60.5±2.9
RAM (ours)	CLIP+DINOv2	5.9±0.3	47.1±1.4	56.5±2.7
RAM-APL (ours)	CLIP+DINOv2	6.5±0.4	47.5±1.9	58.7±2.2

Benchmark Comparison: Food-101

- We improve Top-1 over Random by an average of +4.44%.
- We consistently outperform strong baselines such as GC [Wei13], MIN [Snell17], and Forgetting [Toneva19].
- We scale to large fine-grained datasets with strong gains at low budgets.

Table: Table 9. Comparison with baselines on Food-101. "Traditional" stands for models pretrained on Food-101 for ten epochs. Means and variances are across three runs. Best in bold; second-best in blue. Tags indicate sources.

Methods	IE	1%	10%	30%	50%	70%	Avg. vs. Random
Random	-	6.8±0.4	45.5±0.5	69.5±0.2	76.2±0.3	79.6±0.3	-
Herding [Chen12]	Traditional	6.6±0.5	42.6±1.2	64.6±0.2	74.3±0.2	78.7±0.3	-2.16
KCG [Sener18]		4.7±0.3	35.0±0.7	67.2±0.2	75.9±0.2	79.6±0.2	-3.04
CD (cluster diversity) [MacQueen67]		3.9±0.1	27.7±0.6	66.2±0.4	75.7±0.2	80.0±0.1	-4.82
Margin [Settles10]		5.0±0.7	36.8±2.7	68.4±0.3	76.4±0.1	79.8±0.2	-2.24
Forgetting [Toneva19]		9.5±0.5	53.4±0.5	71.7±0.1	77.3±0.1	79.9±0.1	+2.84
GraNd [Paul21]		2.9±0.2	18.4±0.4	59.7±0.5	74.7±0.2	80.0±0.1	-8.38
CAL/CRAIG [Mirzasoleiman20]		10.3±0.4	51.8±0.3	69.5±0.5	76.0±0.1	79.5±0.2	+1.90
Glister [Killamsetty21]		6.9±0.3	38.8±1.1	67.4±0.4	75.9±0.4	80.1±0.2	-1.70
GC [Wei13]		10.8±0.4	54.4±0.3	71.3±0.4	76.8±0.2	79.5±0.3	+3.04
MDS [Kennard69]		6.1±0.5	45.1±0.4	69.7±0.3	76.2±0.2	79.6±0.1	-0.18
MIN [Snell17]		9.4±0.2	52.1±0.6	70.7±0.7	76.7±0.1	79.8±0.1	+2.22
Ours (RAM-APL)	CLIP+DINOv2	12.3±0.3	56.1±0.6	72.3±0.1	77.9±0.1	81.2±0.2	+4.44

Baselines: Herding [Chen12]; K–Center Greedy/CoreSet [Sener18]; GC (Graph Cut) [Wei13]; MDS (Kennard–Stone) [Kennard69]; CD (k-means cluster distance) [MacQueen67]; Margin

Fusion Matters: Ranking-Space Fusion vs. Concatenation

- We find that simple concatenation underperforms our RAM-APL ranking-space fusion.
- We observe that ranking-space fusion is robust to FM misalignment and scale differences.
- We see consistent benefits across sampling rates.

Table: Table 3. Comparison of feature fusion strategies (Top-1 accuracy, mean±std).

Fusion strategy	1%	10%	30%	50%	70%
Concatenate	5.9±0.4	16.3±0.4	31.7±1.3	47.7±3.0	57.8±1.2
Ours (RAM-APL)	6.5±0.4	15.2±1.2	32.4±2.9	47.5±1.9	58.7±2.2

Generalization Across Target Architectures

- We show that selected subsets transfer to different target models (e.g., MBV3).
- We observe that RAM-APL surpasses baselines even when the IE differs from the target.
- We support practical, model-agnostic subset selection.

Table: Table 6. Cross-architecture generalization of RAM-APL. Target model: MobileNet-V3 (MBV3).

Method	IE → Target Model	10%	30%	50%
Random	MBV3 → MBV3	10.9±1.1	42.1±3.6	61.6±1.9
Forgetting [Toneva19]	MBV3 → MBV3	13.3±0.8	42.0±2.0	61.0±2.3
GC [Wei13]	MBV3 → MBV3	12.4±1.7	40.4±0.2	61.3±1.5
MDS [Kennard69]	MBV3 → MBV3	11.9±0.7	39.8±2.0	62.1±3.3
MIN [Snell17]	MBV3 → MBV3	11.9±1.8	38.4±0.8	61.0±1.4
RAM-APL (Ours)	(CLIP+DINOv2)→ MBV3	13.6±0.3	45.7±0.9	62.3±1.4

Baselines: Herding [Chen12]; K–Center Greedy/CoreSet [Sener18]; GC (Graph Cut) [Wei13]; MDS (Kennard–Stone) [Kennard69]; CD (k-means cluster distance) [MacQueen67]; Margin

Parameter Study: Adaptive Weighting Works Best

- We find that setting alpha to 0.2 and beta to 1 outperforms an equal-weight fusion.
- We support a curriculum-style weighting that increases the emphasis on ambiguity as the budget grows.
- We observe robust default behavior across datasets.

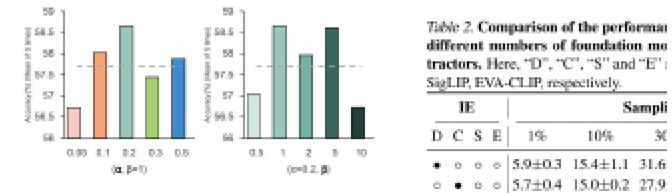

Figure: Parameter analysis at 70% sampling on Pet: our scheduled fusion outperforms an equal-weight, no-schedule fusion of RAM and the complement of APL.

Limitations and When FMs Help Less

- We see limited advantage on noisy, coarse-grained datasets (e.g., CIFAR-10N).
- We note that FM choice depends on dataset, budget, and selector—there is no single winner.
- We may risk overlooking rare-but-important cases without explicit fairness or coverage constraints.

Future Work & Broader Impact

- We plan automatic FM weighting/selection and better fusion for coarse or noisy settings.
- We aim to extend beyond images (e.g., audio, video, multimodal) and beyond classification.
- We will integrate robustness and fairness constraints to protect minorities and rare cases.
- We enable scalable training on large or evolving datasets with lower cost.

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Open Questions

- How can we automatically select and weight FMs for a given dataset and budget?
- Can we extend RAM-APL beyond images (e.g., audio, video, multimodal) and beyond classification?
- How do we make selection robust to extreme label noise, class imbalance, and domain shift?
- Can we enforce fairness or coverage constraints to protect rare subpopulations without hurting accuracy?
- What are the best low-cost proxies for FM complementarity when scaling to many FMs?